23 RADIATION AND HALF-LIFE

In this chapter you will learn about ways of detecting radiation, and where some of the radiation around us comes from. You will also learn why we use a value called half-life to describe the activity of radioactive isotopes.

LEARNING OBJECTIVES

- Know that photographic film or a Geiger–Müller detector can detect ionising radiation
- Explain the sources of background (ionising) radiation from Earth and space
- Know that the activity of a radioactive source decreases over time and is measured in becquerels
- Know the definition of the term half-life and understand that it is different for different radioactive isotopes
- Use the concept of the half-life to carry out simple calculations on activity, including graphical methods

DETECTING IONISING RADIATION

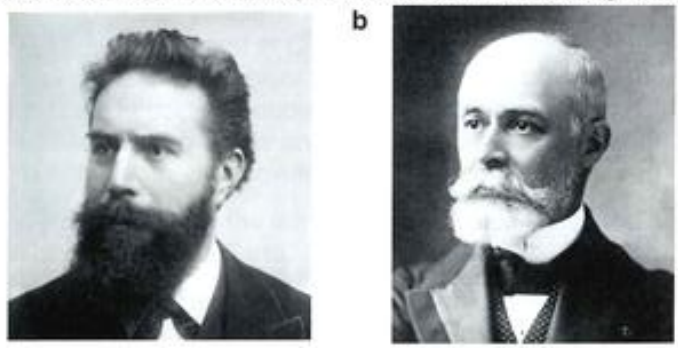
USING PHOTOGRAPHIC FILM

Wilhelm Röntgen (Figure 23.1a) discovered x-rays in 1895. Henri Becquerel (Figure 23.1b) believed that uranium emitted x-rays after being exposed to sunlight. To test this idea he placed some wrapped, unused photographic plates in a drawer with some samples of uranium **ore** on top of them. He found a strong image of the ore on the plates when he developed them. He later realised that this was due to a new type of ionising radiation. He had discovered radioactivity.

The unit of radioactivity is named after Becquerel. The becquerel (Bq) is a measure of how many unstable nuclei are disintegrating (breaking up) per second – one becquerel means a rate of one disintegration per second. The becquerel is a very small unit. More practical units are the kBq (an average of 1000 disintegrations per second) and the MBq (an average of 1 000 000 disintegrations per second).

Photographic film can still be used to detect radioactivity. Some scientists who work with radioactive materials wear a piece of photographic film in a badge. If the film becomes fogged (unclear) it shows that the scientist has been exposed to a certain amount of radiation. These badges are checked regularly to ensure that scientists are not exposed to too much ionising radiation.

a b

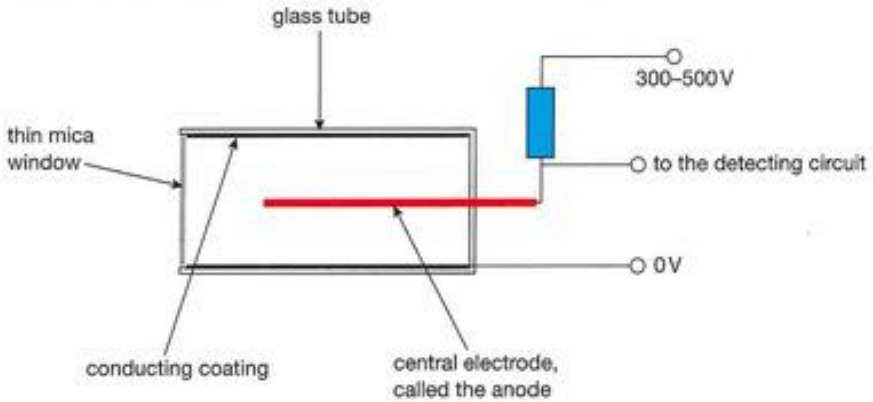


▲ Figure 23.1 a Wilhelm Röntgen (1845–1923), b Henri Becquerel (1852–1908). Nobel prize winners in 1901 and 1903

THE GEIGER–MÜLLER (GM) TUBE

Figure 23.2 shows the basic construction of a Geiger–Müller (GM) tube. It is a glass tube with an electrically conducting surface on the inside. The tube has a thin window made of mica (a naturally occurring mineral that can be split into thin sheets). The tube contains a special mixture of gases at very low pressure. In the middle of the tube, electrically insulated from the conducting surface, there is an **electrode**. This electrode is connected, via a high-value resistor, to a high-voltage supply, typically 300–500 V.

When ionising radiation enters the tube it causes the low pressure gas inside to form ions. The ions allow a small amount of current to flow from the electrode to the conducting layer. This is detected by an electronic circuit.

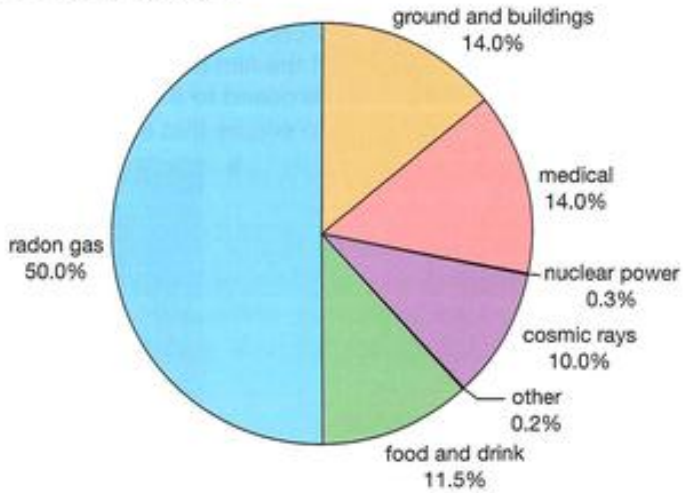


▲ Figure 23.2 A Geiger–Müller tube is used to measure the level of radiation.

The GM tube is usually linked up to a counting circuit. This keeps a count of how many ionising particles (or how much γ radiation) have entered the GM tube. Sometimes GM tubes are connected to rate meters. These measure the number of ionising events per second, and so give a measure of the radioactivity in becquerels. Rate meters usually have a loudspeaker output so the level of radioactivity is indicated by the rate of clicks produced.

BACKGROUND RADIATION

Background radiation is low-level ionising radiation that is produced all the time. This background radiation has a number of sources. Some of these are natural and some are artificial.



▲ Figure 23.3 Sources of background radiation in the UK. These are the average values – the true amounts and proportions vary from place to place.

NATURAL BACKGROUND RADIATION FROM THE EARTH

KEY POINT

When an atom of a radioactive element decays it gives out radiation and changes to an atom of another element. This may also be radioactive, and decay to form an atom of yet another element. The elements formed as a result of a radioactive element experiencing a series of decays are called decay products.

Some of the radiation we receive comes from rocks in the Earth's crust (hard outer layer). When the Earth was formed, around 4.5 billion years ago, it contained many radioactive isotopes. Some decayed very quickly but others are still producing radiation. Some of the decay products of these long-lived radioactive materials are also radioactive, so there are radioactive isotopes with much shorter half-lives (see page 237) still present in the Earth's crust.

One form of uranium is a radioactive element that decays very slowly. Two of its decay products are gases. These are the radioactive gases radon and thoron. Radon-222 is a highly radioactive gas produced by the decay of radium-226. Thoron, or radium-220, is an isotope of radium formed by the decay of a radioactive isotope of thorium (thorium-232).

As these decay products are gases, they come out of radioactive rocks. They are dense gases so they build up in the basements of buildings. Some parts of the Earth's crust have higher amounts of radioactive material so the amount of background radiation produced in this way varies from place to place. In Cornwall in the UK, for example, where the granite rock contains traces of uranium, the risk of exposure to radiation from radon gas is greater than in some other parts of the UK.

NATURAL BACKGROUND RADIATION FROM SPACE

Violent nuclear reactions in stars and exploding stars called supernovae produce cosmic rays (very energetic particles) that continuously hit the Earth. Lower energy cosmic rays are given out by the Sun. Our atmosphere gives us fairly good protection from cosmic rays but some still reach the Earth's surface.

RADIATION IN LIVING THINGS

The atoms that make up our bodies were formed in the violent reactions that take place in stars that exploded (supernovas) billions of years ago. Some of these atoms are radioactive so we carry our own personal source of radiation around with us. Also, as we breathe we take in tiny amounts of the radioactive isotope of carbon, carbon-14. Because carbon-14 behaves chemically just like the stable isotope, carbon-12, we continuously renew the amount of the radioactive carbon in our bodies (see page 245).

Carbon-14 and other radioactive isotopes are eaten by humans (and animals which are in turn eaten by humans) because they are present in all living things.

ARTIFICIAL RADIATION

We use radioactive materials for many purposes. Generating electricity in nuclear power stations has been responsible for the leaking of radioactive material into the environment. The levels are usually small, but there have been a number of major incidents around the world, especially at Three Mile Island in the USA in 1979 and at Chernobyl in the Ukraine in 1986. The tsunami and earthquake that caused major damage and loss of life in Japan in 2011 also damaged the Fukoshima nuclear power station resulting in the release of radioactive materials into the air and the ocean as well as making a large area of land around the damaged power station unsafe for humans to live there.

Testing nuclear weapons in the atmosphere has also increased the amounts of radioactive isotopes on the Earth.

Radioactive **tracers** are used in industry and medicine. Radioactive materials are also used to treat certain forms of cancer. However, the majority of background radiation is natural – the amount produced from medical use in industry is very small indeed.

RADIOACTIVE DECAY

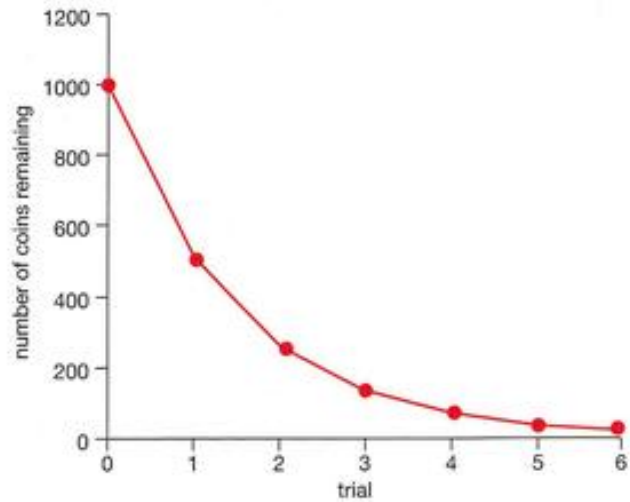
Radioactive decay is a random (unpredictable) process, just like throwing a coin. If we throw a coin we cannot say with certainty whether it will come down heads or tails. If we throw a thousand coins we cannot predict which will land heads and which will land tails. The same is true for radioactive nuclei. It is impossible to tell which nuclei will disintegrate (break down) at any particular time. However, if we threw a thousand coins we would be surprised if the number that landed as heads was not around 500. We know that a normal coin has an equal chance of landing as a head or a tail, so if we got 600 heads we would think it was unusual. If the proportion of heads were much greater than this we would be right to think that the coin was not fair.



▲ Figure 23.4 Throwing a coin

EXPERIMENTAL DEMONSTRATION OF NUCLEAR DECAY

We could, if we had the time, take 1000 coins and throw them. We could then remove all the coins that came down heads, note the number of coins remaining and then repeat the process. If we did this for, say, six trials we would begin to see the trend. A set of typical results is shown in the following table and in Figure 23.5.



▲ Figure 23.5 Coin-throwing experiment. Each time the coins are thrown about 50% of them land as 'heads' and are removed from the pile. The graph decreases steeply at first but then does so more and more slowly.

Trial	Number of coins remaining
0	1000
1	519
2	264
3	140
4	72
5	33
6	19

EXTENSION WORK

The coin-throwing experiment is a model of radioactive decay. A good model will show the features of the real process. We must remember that models have limitations, however, and do not perfectly represent the actual process. One limitation of the experiment is that of scale. In just 1 g of uranium-235 there are millions of nuclei, and our model uses just 1000 coins. The model would be better if we used, say, 1 000 000 coins, but would take too long to perform. If we use a computer model to throw the coins we could deal with more realistic numbers.

Notice that the graph in Figure 23.5 falls steeply at first and more slowly after each throw. How quickly the graph falls depends on how many heads occur on each throw. But as the number of coins decreases, the number of coins that come up heads also gets smaller. This graph follows a rule: the smaller the quantity, the more slowly the quantity decreases. The quantity here is the number of coins still in the experiment. The name for this kind of decrease proportional to size is called exponential decay.

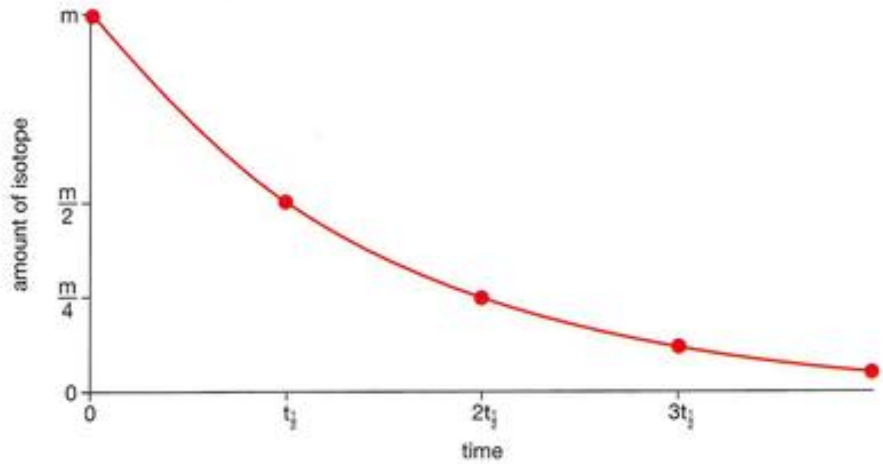
If we have a sample of a radioactive material, it will contain millions of atoms. The process of decay is random, so we don't know when an atom will decay but there will be a probability that a certain fraction of them will disintegrate in a particular time. This is the same as in the coin toss – there was a 50% probability that the coins would land heads each time we conducted a trial. Once an unstable nucleus has disintegrated, it is out of the game – it won't be around to disintegrate during the next period of time. If we plot a graph of number of disintegrations per second against time for a radioactive isotope we would, therefore, expect the rate of decay to fall as time passes because there are fewer nuclei to decay.

HALF-LIFE

Our coin-tossing model of radioactive decay shows a graph that approaches the horizontal axis more and more slowly as time passes. The model will produce a number of throws after which all the coins have been taken out of the game. The number is likely to vary from trial to trial because the model becomes less and less reliable as the number of coins becomes smaller. With real radioactive decay we use a measure of activity called the **half-life**. This is defined as follows.

The half-life of a radioactive sample is the average time taken for half the original mass of the sample to decay. If the amount of radioactive matter has halved then the activity of the decay halves – this activity is what is measured in finding the half-life of an isotope. The half-life is different for different radioactive isotopes.

Figure 23.6 shows what this means. After one half-life period, $t_{\frac{1}{2}}$, the amount of the original unstable element has halved. After a second period of time, $t_{\frac{1}{2}}$, the amount has halved again, and so on.



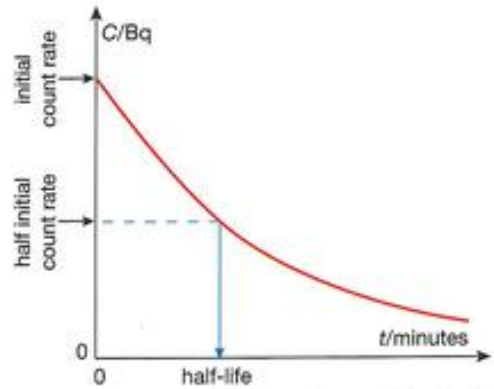
▲ Figure 23.6 Graph showing the half-life period for a radioactive isotope

MEASURING THE HALF-LIFE OF A RADIOACTIVE ISOTOPE

To measure the half-life of a radioactive material (**radioisotope**) we must measure the activity of the sample at regular times. This is done using a Geiger–Müller (GM) tube linked to a rate meter. Before taking measurements from the sample, we must measure the local background radiation. We must **subtract** the background radiation measurement from measurements taken from the sample so we know the radiation produced by the sample itself. We then measure the rate of decay of the sample at regular time intervals. The rate of decay is shown by the count rate on the rate meter. The results should be recorded in a table like the one shown below.

Average background radiation measured over 5 min = x Bq		
Time, *t*/min	Count rate/Bq	Corrected count rate, *C*/Bq
0	y_0	$y_0 - x$
5	y_5	$y_5 - x$

The rate of decay, *C*, corrected for background radiation, is proportional to the amount of radioactive isotope present. If we plot a graph of *C* against time, *t*, we can measure the half-life from the graph, as shown in Figure 23.7.



▲ Figure 23.7 You can find the half-life by reading from the graph the time taken for the count rate to halve.

As we have already mentioned, different isotopes can have very different half-lives. Some examples of different half-lives are shown in the table below.

Isotope	Half-life	Decay process
uranium-238	4.5 billion years	α particle emission
radium-226	1590 years	α particle emission, γ ray emission
radon-222	3.825 days	α particle emission

Isotopes with short half-lives are suited to medical use (see page 241). This is because the activity of a source will rapidly become very small as the isotope decays quickly.

Isotopes used for dating samples of organic material need to have very long half-lives. This is because the activity will become difficult to measure accurately if it drops below a certain level. In Chapter 24, we shall see that there are suitable isotopes for these different applications.

HALF-LIFE CALCULATIONS

Graphs of activity, in becquerels, against time can be used to find the half-life of an isotope, and this half-life information can be used to make predictions of the activity of the radioisotope at a later time.

EXAMPLE 1

The activity of a sample of a certain isotope is found to be 200 Bq.

If the isotope has a half-life of 20 minutes, what will the activity of the sample be after one hour?

After 20 minutes the activity will have halved to 100 Bq.

After 40 minutes (two half-lives) it will have halved again to 50 Bq. After 60 minutes it will have halved again, so the activity will be 25 Bq.

What is the level of activity of this sample after three hours?

Three hours = 9 × 20 minutes – that is, nine half-life periods. This means the activity will have halved nine times. The level of activity (and the amount of the radioisotope remaining) will be:

$$\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}$$

or $\frac{1}{2^9}$ of the original value, and so $\frac{1}{512}$ of the original activity or amount.

CHAPTER QUESTIONS

More questions on radioactive decay can be found at the end of Unit 7 on page 255.

SKILLS CRITICAL THINKING

9 1 a Explain what is meant by background radiation.

b Explain the difference between natural background radiation and artificial background radiation.

8 c Give three different sources of background radiation. Say whether your examples are natural or artificial sources.

SKILLS CRITICAL THINKING

2 a Explain simply the principle of the Geiger–Müller tube.

b The Geiger–Müller tube is often connected to a rate meter. Explain what this instrument measures.

c The rate meter is calibrated in kBq. How is this unit defined?

3 a Define what is meant by the half-life of a radioactive material.

b Radioactive decay is a random process. Explain what this means.

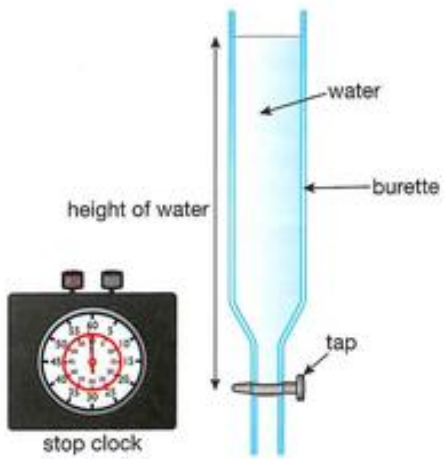
SKILLS PROBLEM SOLVING

4 The activity of a radioactive sample is measured. The activity, corrected for background radiation, is found to be 240 Bq. The activity is measured again after 1 hour 30 minutes and is now 30 Bq. What is the half-life of the sample?

5 In another model of radioactive decay, a student fills a burette with water, as shown in the diagram, and starts a timer at the instant the tap at the bottom is opened. She notes the height of the column of water at regular intervals. It takes 35 seconds to empty from 50 ml to 25 ml. Assuming that the arrangement provides a good model of radioactive decay:

a How long will it take for three quarters of the water in the burette to drain away?

b How much water should remain in the burette after 1¾ minutes?



SKILLS EXECUTIVE FUNCTION

6 A student wants to measure the half-life of a radioactive isotope. He is told the isotope has a half-life of between 10 and 20 minutes. Illustrating your answers as appropriate, describe:

a the measurements that he should take

b how he should use the measurements to arrive at an estimate of the half-life for the isotope.